

Metaphysical Foundations and Triangulation Invariants of the Neurosymbolic Forest

System Architecture Invariant Manifest

September 25, 2026

1 Introduction and Ontological Modalities

Following the formalisms of Edward Zalta’s Abstract Object Theory and Situation Theory, we isolate three distinct modal world-lines within a single unified processing manifold:

1. **The Historical Immutable World:** Rigidly designated actualities (s_{actual}) frozen in the past (e.g., Moses, David, Paul). They are structurally locked against retroactive modifications.
2. **The Fictional Mutable World:** Abstract objects encoding properties inside an isolated situation domain (s'), completely open to counterfactual planning rewrites (e.g., John, Judith, Pierre).
3. **The Indexical Real World:** The concrete evaluation layer where sovereign actions and executions occur in real time.

2 The DSL Discretization Interface and Homomorphisms

Continuous environments (e.g., physics grids, maze navigation, or real-time spatial physics) present an infinite, unsearchable behavior space. We address this challenge by introducing an intermediate LISP-like Domain-Specific Language (DSL) that functions as a structural discretization filter.

2.1 When the DSL and Trees Activate

The runtime execution cascade operates in a strict top-down dependency chain:

$$\text{Continuous Simulation} \xrightarrow{\text{DSL Filter}} \text{Discrete S-Expressions} \xrightarrow{\text{FCA Layout}} \text{Decision Tree Leaf Nodes} \quad (1)$$

The DSL is active at the very beginning of the loop. It ingests continuous floating-point observations and maps them into low-dimensional atomic sensor values (`raydist`, `distance`, `gap`). These expressions are then feature-engineered into discrete attribute profiles for our Decision Trees.

2.2 Homomorphism as Terminal Node Agreement

Under Jon Barwise’s Information Flow framework, an analogy is defined as a structure-preserving channel infomorphism (homomorphism). In a Decision Tree model, a homomorphism is achieved when *two completely different problems reach the exact same terminal leaf node* through feature engineering. By sharing a common destination vertex, the system registers a verified logical alignment across separate world-lines.

2.3 The Execution Hand-Off to Self-Similar Groups

Once the Decision Tree categorizes a situation at its terminal leaf node, *the self-similar group automaton takes over as the execution motor*. Rather than consulting brittle, static tables of connections like Newell’s General Problem Solver (GPS), the terminal node maps the problem directly to recursive algebraic generators (a, b, c, d). The path is discovered dynamically by solving the algebraic Word Problem on a symmetric rooted tree, collapsing safely to the identity element (e) upon completion.

3 The Core Logic of the Unified Background Knowledge Base (`kb.pl`)

Rather than forcing individual decision trees to duplicate rules, our architecture centralizes all invariant logical axioms inside a single background knowledge layer: `kb.pl`.

3.1 Pragmatic Contract Governance

`kb.pl` handles second-order social contract permissions. Given an action p and a mandatory authorization constraint q , an unauthorized exception state triggers an immediate structural breach vertex:

$$\text{Pragmatic_Breach}(p, q) \iff p \wedge \neg q \quad (2)$$

3.2 Analogy Fact Transfer Laws

The centralized engine hosts the rules of logical transfer:

1. **Reflexive Rule:** Symmetrically infers facts between separate domains ($S_1 \neq S_2$) if an underlying structural match is satisfied.
2. **Transitive Rule:** Chains recovery metrics across multi-point domains ($S_1 \rightarrow S_2 \rightarrow S_3$) by tracing back to the root fact origin point.

4 The Principle of Triangulation via Channel Isomorphisms

While Jon Barwise formulated Channel Composition to track how information commutes through distributed networks, we deploy the Principle of Triangulation to secure cross-model analogies. If three completely independent structural representations (FCA Parthood, Graph Topologies, and Fallacy Patterns) all arrive at the exact same leaf category, the analogy is mathematically finalized.

5 Contravariant Lifecycles: Monkey and Experimenter

In Tree #11 (Experiential Architectural Memory), the system learns to parse contravariant observational learning tasks. While the Monkey-and-Bananas problem represents an extraction phase and the Experimenter-and-Bananas problem represents a setup phase, they share an identical topological invariant pattern executed in reverse temporal order. By feature-engineering their shared algebraic operations, the tree maps both settings to a unified strategy leaf vertex: `homomorphic_imitation_learning`, proving that the transition from setting up a constraint to solving it is governed by an invariant top-down law of redemption.